Lecture 01: Introduction to Database Management Systems

ITCS216 Fundamentals of Database Systems
Semester 1/2026

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Course Learning Outcome (CLOs)

After this courses, students are able to:

- **CLO1**: **Explain** the concepts and components of database and the database management system.
- **CLO2**: **Analyze** the given real world problems' requirements in order to model it using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
- **CLO3: Demonstrate** an understanding of the relational data model.
- **CLO4**: **Formulate** SQL command to create tables and indexes, insert/update/delete data, and query data in a relational DBMS.

Preparation

Resources

Database Design Tool



Download: https://dev.mysql.com/downloads/workbench/



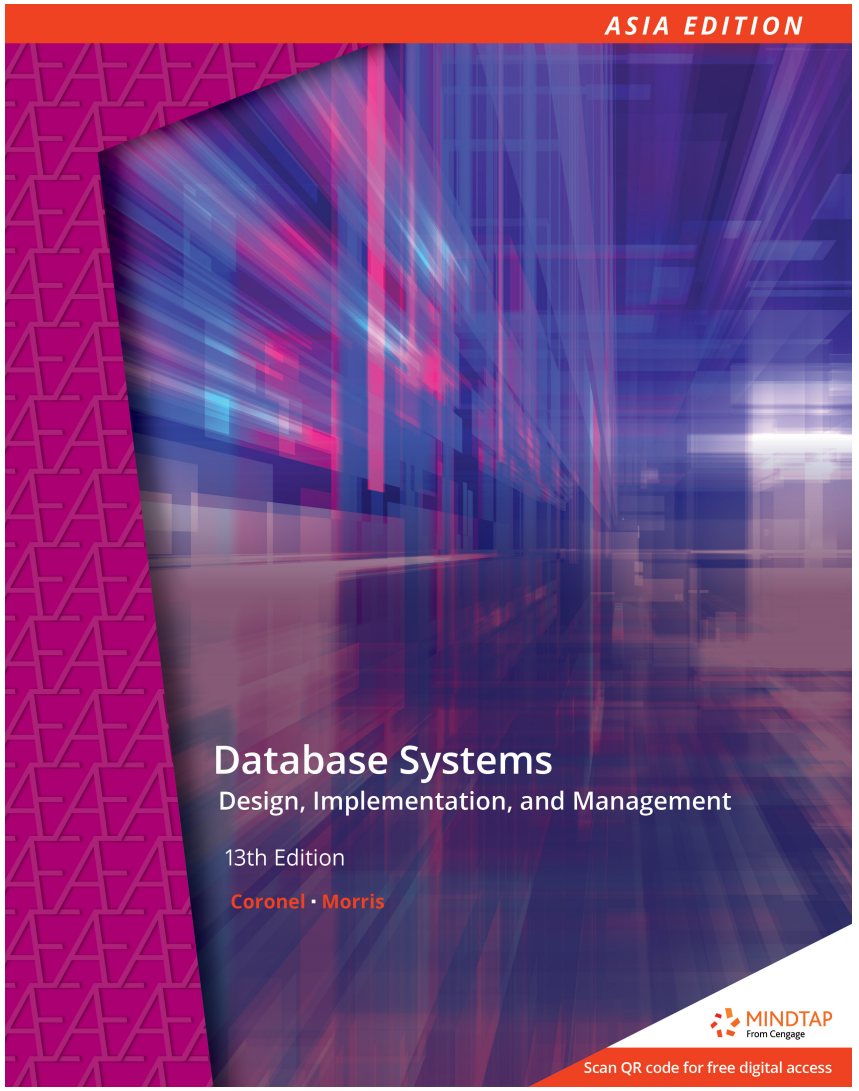
Draw.io

ONLINE: https://app.diagrams.net/

OFFLINE: https://github.com/jgraph/drawiodesktop/releases/tag/v21.6.5)

Textbook

Database Systems: Design, Implement, and Management



How to get A

Assessment and Grading Criteria

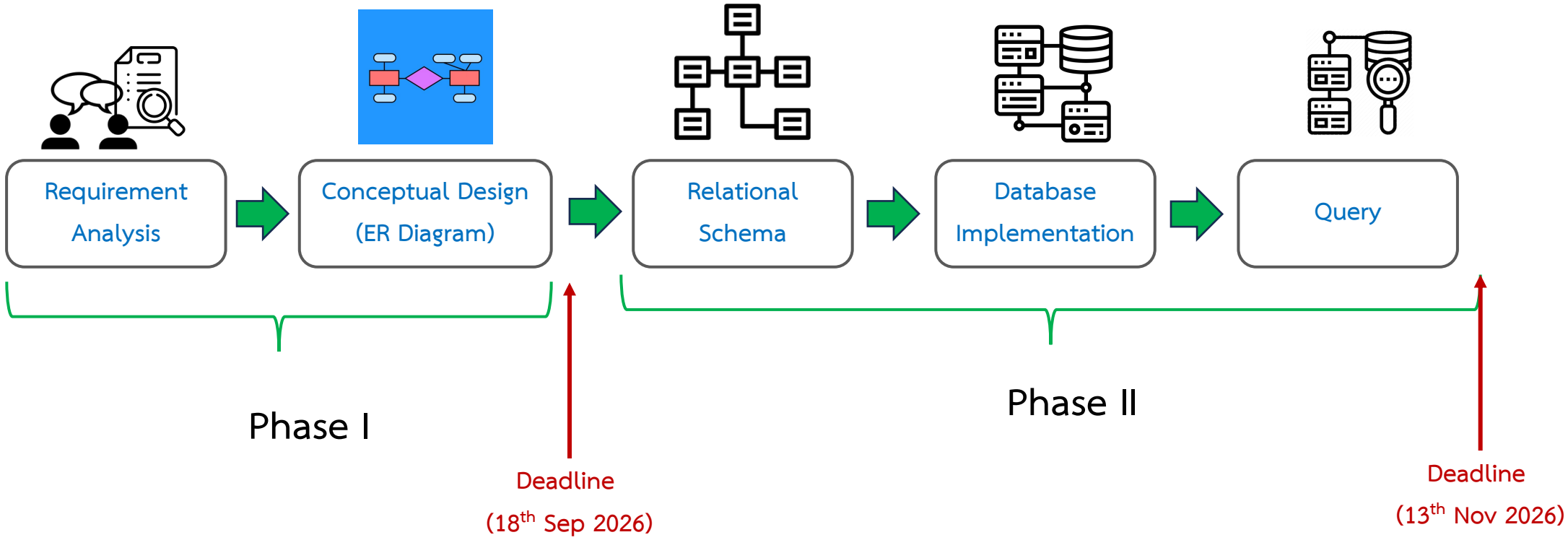
- **Participation** **(10%)**

- **Group Project** **(40%)**
  - Project phase I 15%
  - Project phase II 15%
  - Project Exam 10%
- **Examination** **(50%)**
  - Midterm exam (paper-based) 25%
  - Final exam (Paper-based and Finex system) 25%

DBMS Project (40%)

Project Description (Group work)

5 Members



Learning Protocol

Agreement

- **Communications**
  - If it's course-related, use **Mahidol** **e-mails**.
  - Avoid using social media for work-related communication unless it's urgent.

- **Project Consultation**
  - Project consultation will be provided on **Week 3 -7** and **Week 11-14 location is tentatively at Lobby 2nd Floor.** The schedule will be provided on the project overview.
- **Ask away!**
  - There's no such thing as a dumb question! But if you don't ask, you may remain dumb.

AI Policy

According to MU Regulations

- **LEVEL 1: NO AI**
  - You must complete the tasks entirely without the use of Generative AI tools. Rely solely on your own knowledge, understanding, and skills. The goal is to demonstrate your independent thinking and mastery of the subject matter without external AI assistance.

  However, for our class,

  - *Students may use Grammarly for editing project reports.*
  - *For the project implementation, students may use Generative AI to support data generation and debugging. However, students must be able to explain and justify all aspects of their submitted work.*
  - *Students are responsible for declaring their use of AI tools in all required sections.*
  - *Failure to disclose AI usage, or any misuse of AI, may result in no credit for the assignment.*

.

Let’s get started !!!

Lesson Learning Outcome

After this lesson, students are able to:

- Differenciate between the concept of **data** and **information**, file systems and database systems,
- Understand the main components of database,
- Understand the concept of three schema architecture.

Database Topics

Knowing Database and How to Develop Data Model for Business Application PART I

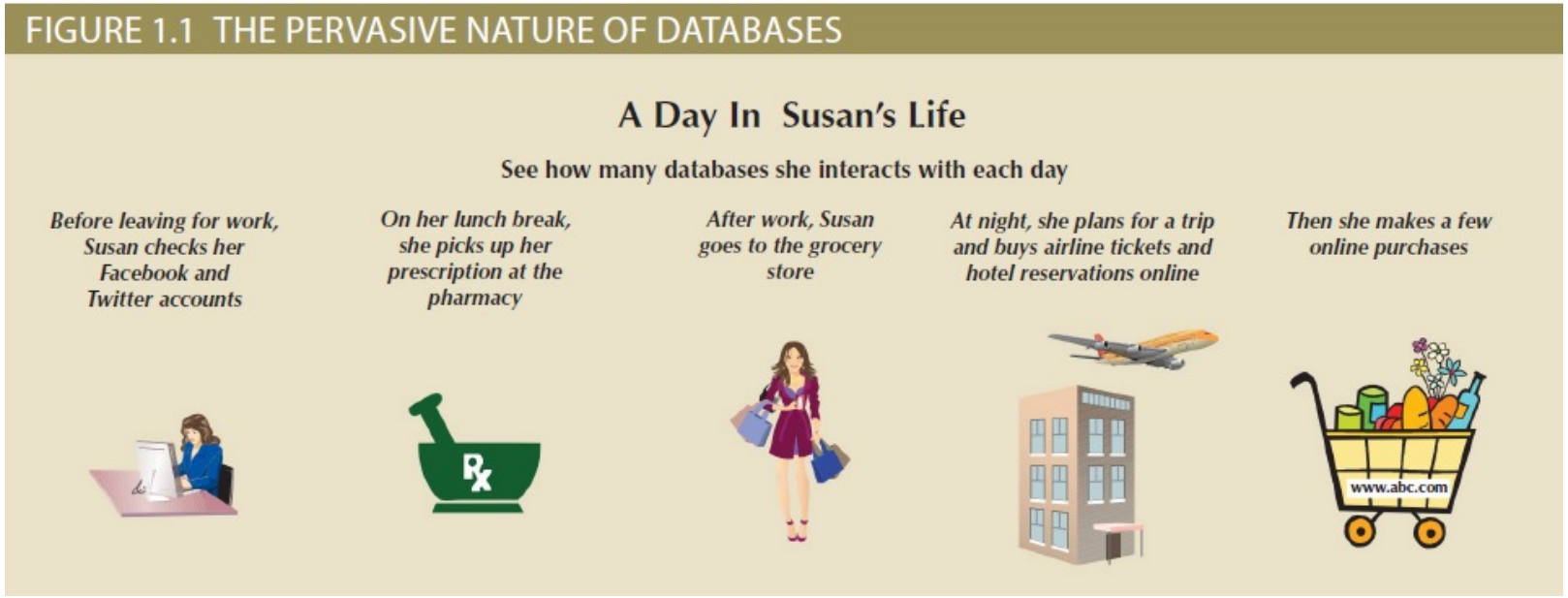
- Fundamental Concepts of DBMS
- The Three-Schema Architecture
- Entity-Relationship Model
- Developing an E-R Diagram
- Relational Data Model and Relational Algebra
- Developing a Relational Database Schema from an E-R Diagram
- Integrity Constraints Basic Concepts

More Inside Database Design, Implementation, and Management PART II

- **SQL Programming**
  - Transaction Management & Concurrency Control
  - Normalization Theory for Relational Database Design
  - Database Design Process
  - Managing Database and SQL Performance.
  - **Databases for Decision Support.**

Data in Everyday life

FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES



Data in Everyday life

Can you identify some of the data generated by your own daily activity?

FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES

A Day In Susan's Life

See how many databases she interacts with each day

Before leaving for work, Susan checks her Facebook and Twitter accounts	*On her lunch break, she picks up her prescription at the pharmacy*	*After work, Susan goes to the grocery store*	*At night, she plans for a trip and buys airline tickets and hotel reservations online*	*Then she makes a few online purchases*
				
Where is the data about the friends and groups stored? Where are the "likes" stored and what would they be used for?	Where is the pharmacy inventory data stored? What data about each product will be in the inventory data? What data is kept about each customer and where is it stored?	Where is the product data stored? Is the product quantity in stock updated at checkout? Does she pay with a credit card?	Where does the online travel website get the airline and hotel data from? What customer data would be kept by the website? Where would the customer data be stored?	Where are the product and stock data stored? Where does the system get the data to generate product "recommendations" to the customer? Where would credit card information be stored?
Users Friends Posts	Products Sales Customers	Products Sales Customers	Flights Hotels Customers	Products Sales Customers

Why Database?

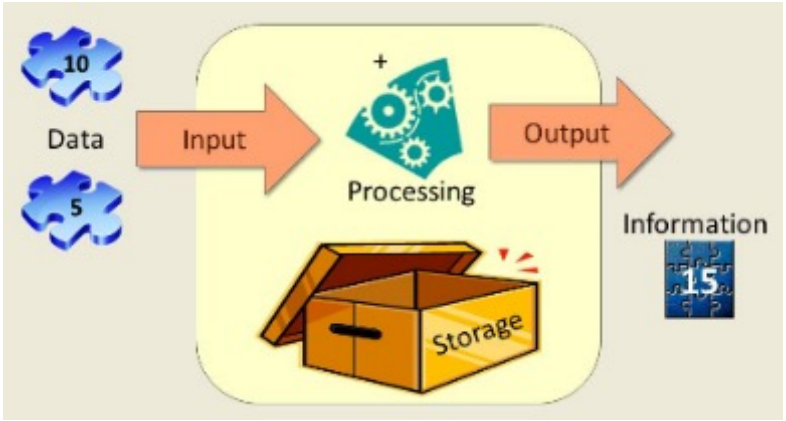
- Databases solve many problems encountered in data management
  - Used in almost all modern settings involving data management, for example:
    - Business
    - Research
    - Administration

- Important to understand how databases work and interact with other applications

Data vs. Information

Data

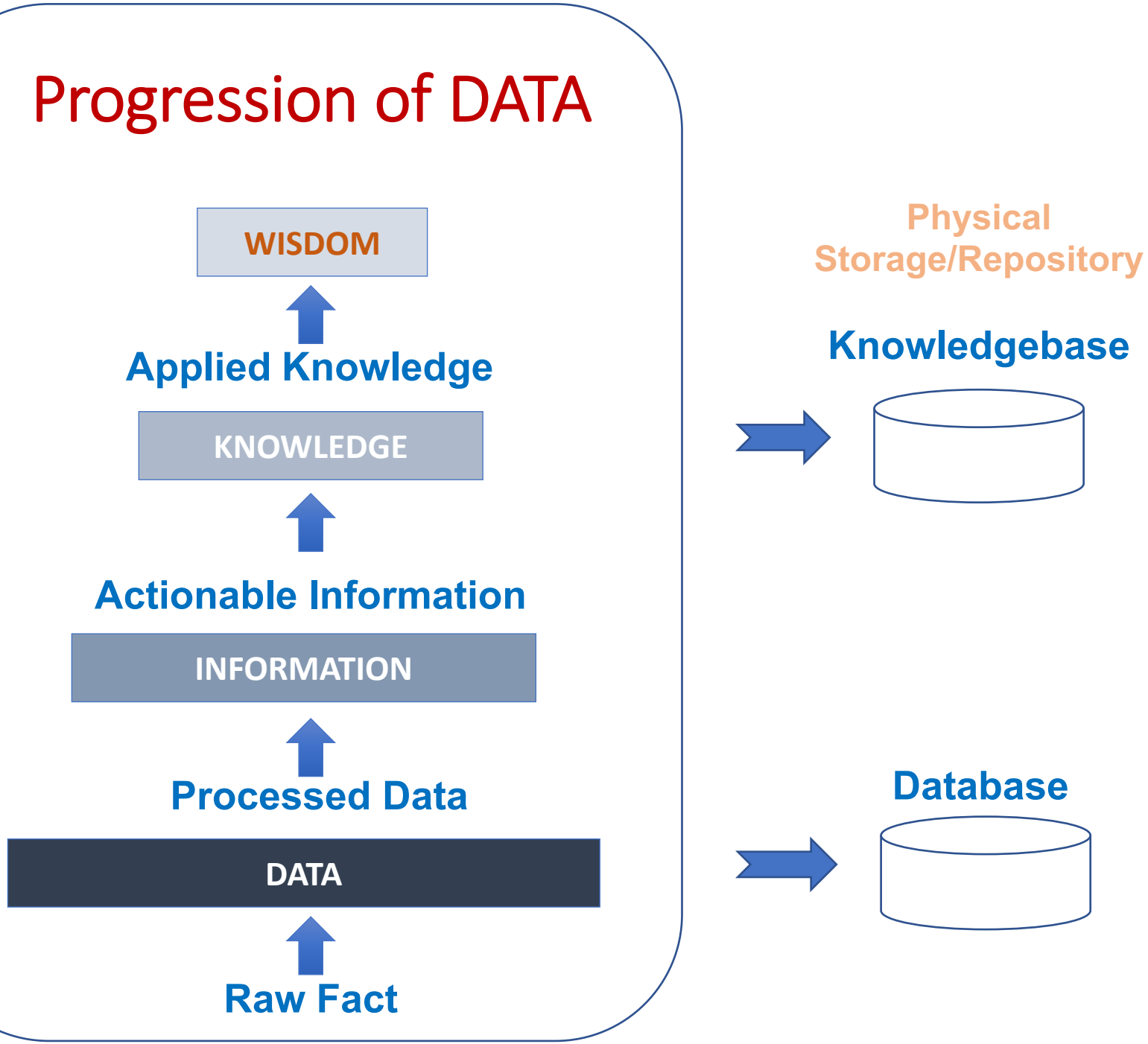
- Raw facts
  - Raw data - Not yet been processed to reveal the meaning
- Building blocks of information
- Data management
  - Generation, storage, and retrieval of data



Information

- Produced by processing data
- Require context to reveal the meaning of data
- Enables **knowledge** creation
- Should be accurate, relevant, and timely to enable good decision making

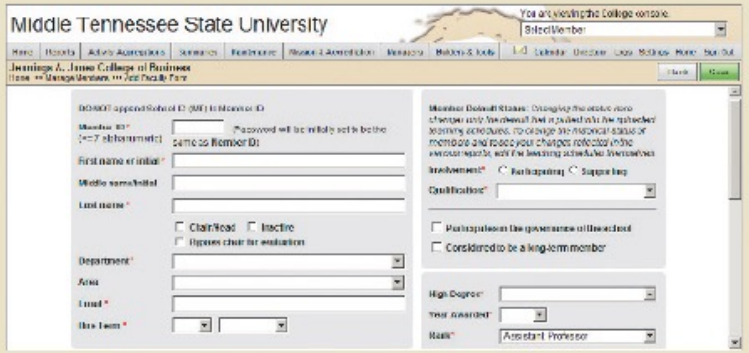
Progression of DATA



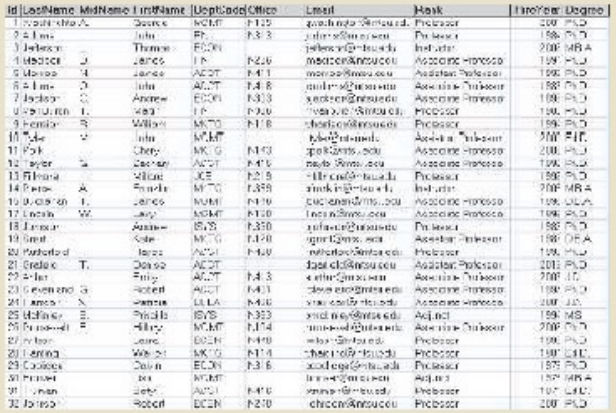
Data vs. Information

FIGURE 1.2 TRANSFORMING RAW DATA INTO INFORMATION

a) Data entry screen



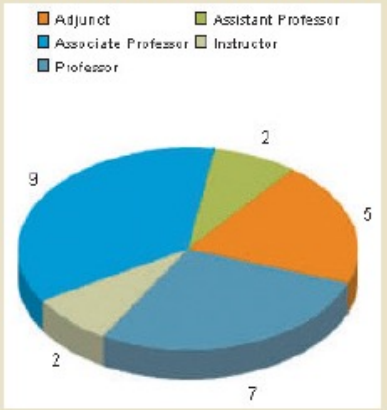
b) Raw data



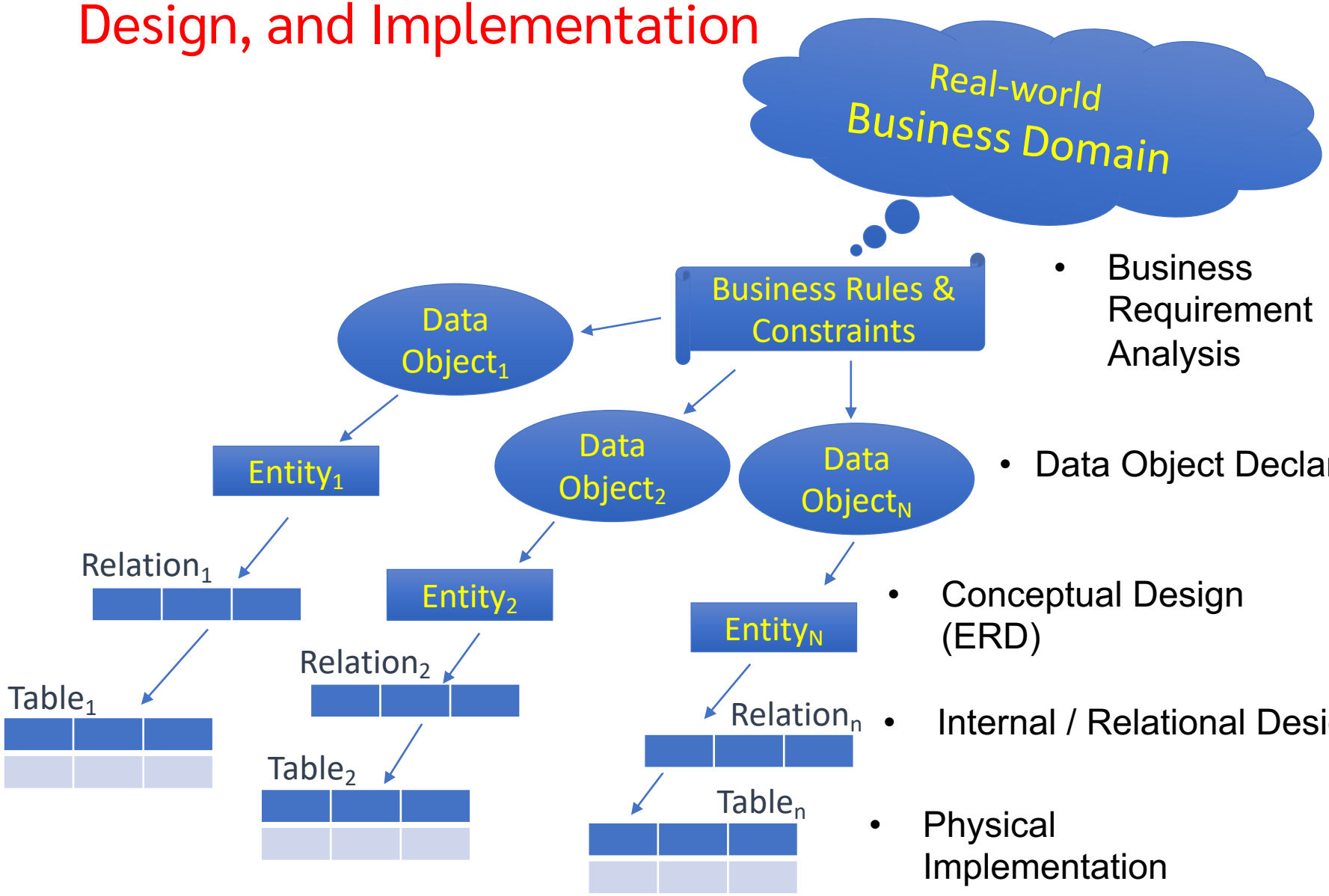
c) Information in summary format

Rank	COUNT	%/INFS	TOT/COL	%/COL. TOT.	%/COL. FAC.
Adjunct	5	20.00%	23	21.74%	3.27%
Assistant Professor	2	8.00%	28	7.14%	1.31%
Associate Professor	9	36.00%	37	24.32%	5.88%
Instructor	2	8.00%	18	11.11%	1.31%
Professor	7	28.00%	47	14.89%	4.58%

d) Information in graphical format



Principle Concept for Database Analysis, Design, and Implementation



- Business Requirement Analysis
- Data Object Declaration
- Conceptual Design (ERD)
- Internal / Relational Design
- Physical Implementation

Principle Concept for Database Analysis, Design, and Implementation

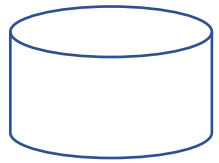
Database Analysis	Database Design	Database Implementation
• Business Domain Understanding • Business Requirements Analysis • Domain Data Object Declaration	• Conceptual Design (ERD) • Relational Schema Mapping	• SQL Development (DDL, DML, QL) • Database Sub-Language Integration
Output	Data Object	Entity	Relation	Table	Physical File
Role	Database Analyst	Database Designer	Database Developer		

Users



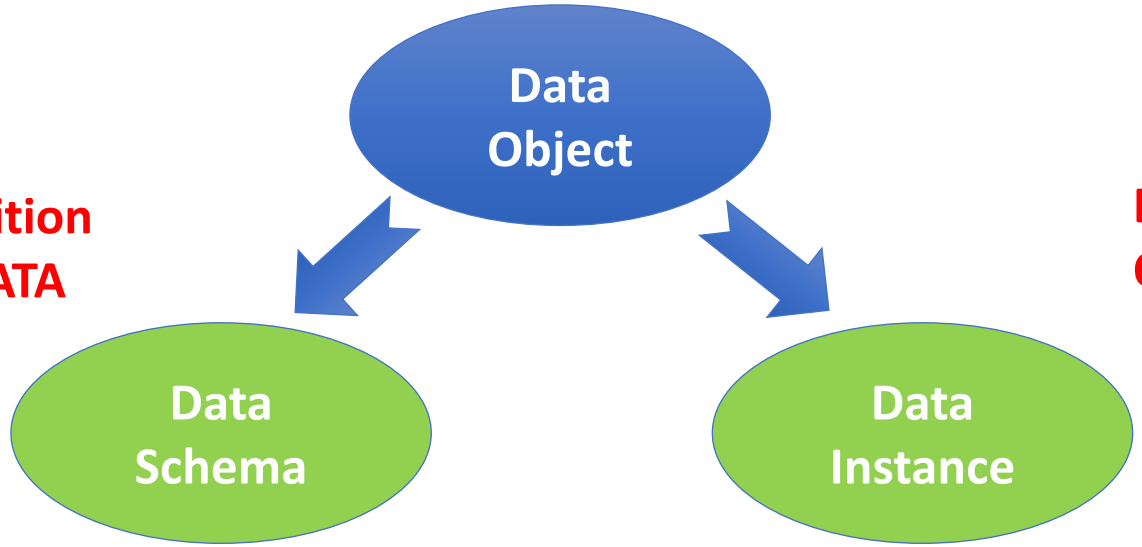
Starting from Users Requirements

To Machine-Processable Data format



Database Repository

Nature of Data Object in Database



Data Definition Or METADATA

Data Instance Or Data Record

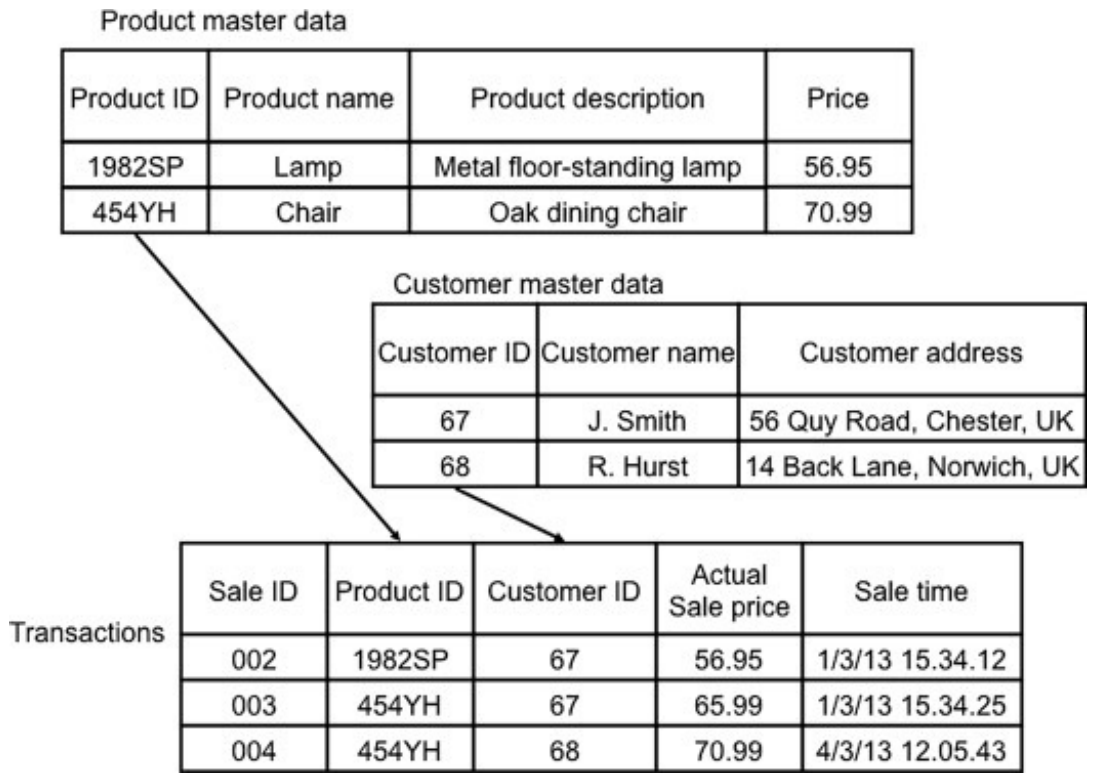
- Skeleton or structure of data
- Properties or Characteristics of data
- Defined as Attribute of data
- For example, Student_Name, Student_ID, Gender

- Raw fact or Raw data
- Structural or Un-structural data
- Defined as Data Record
- For example, Mr.Somboon, 6288000, Male.

Glossary: MetaData is data about data defined the database structure (or database schema).

Master and Transactional Data Object

- Master data object is referred to the data obtained from the main data objects. For example, student, customer, product, or department.
- Transactional data object relates to the transactions produced from business functions (or processes) and included data that is captured, for example, enrollment, order, payment, or reservation.



Product master data

Product ID	Product name	Product description	Price
1982SP	Lamp	Metal floor-standing lamp	56.95
454YH	Chair	Oak dining chair	70.99

Customer master data

Customer ID	Customer name	Customer address
67	J. Smith	56 Quy Road, Chester, UK
68	R. Hurst	14 Back Lane, Norwich, UK

Transactions

Sale ID	Product ID	Customer ID	Actual Sale price	Sale time
002	1982SP	67	56.95	1/3/13 15.34.12
003	454YH	67	65.99	1/3/13 15.34.25
004	454YH	68	70.99	4/3/13 12.05.43



Database Design Issues

Problems with File System Data Processing

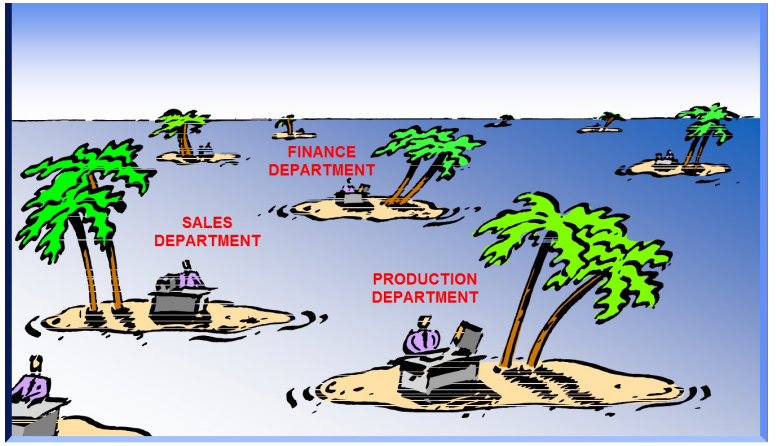
- Summary of file system's limitations:
  - Data retrieval task requires extensive programming,
  - **Ad hoc queries (script language)** is impossible,
  - System administration (create and maintaining number of files) is complex and difficult,
  - Lack of security and data sharing features,
  - Difficult to make changes to existing structures.

Glossary: Ad hoc Query needs when questions arise and are not be able to be solved with predetermined or predefined datasets.

Data Redundancy

- File system structure makes it difficult to combine data from multiple sources:
  - Vulnerable to security breaches
- Organizational structure promotes storage of same data in different locations:
  - Islands of information

- Data stored in different locations is unlikely to be updated consistently
- Data redundancy: same data stored unnecessarily in different places



Data Redundancy (cont'd.)

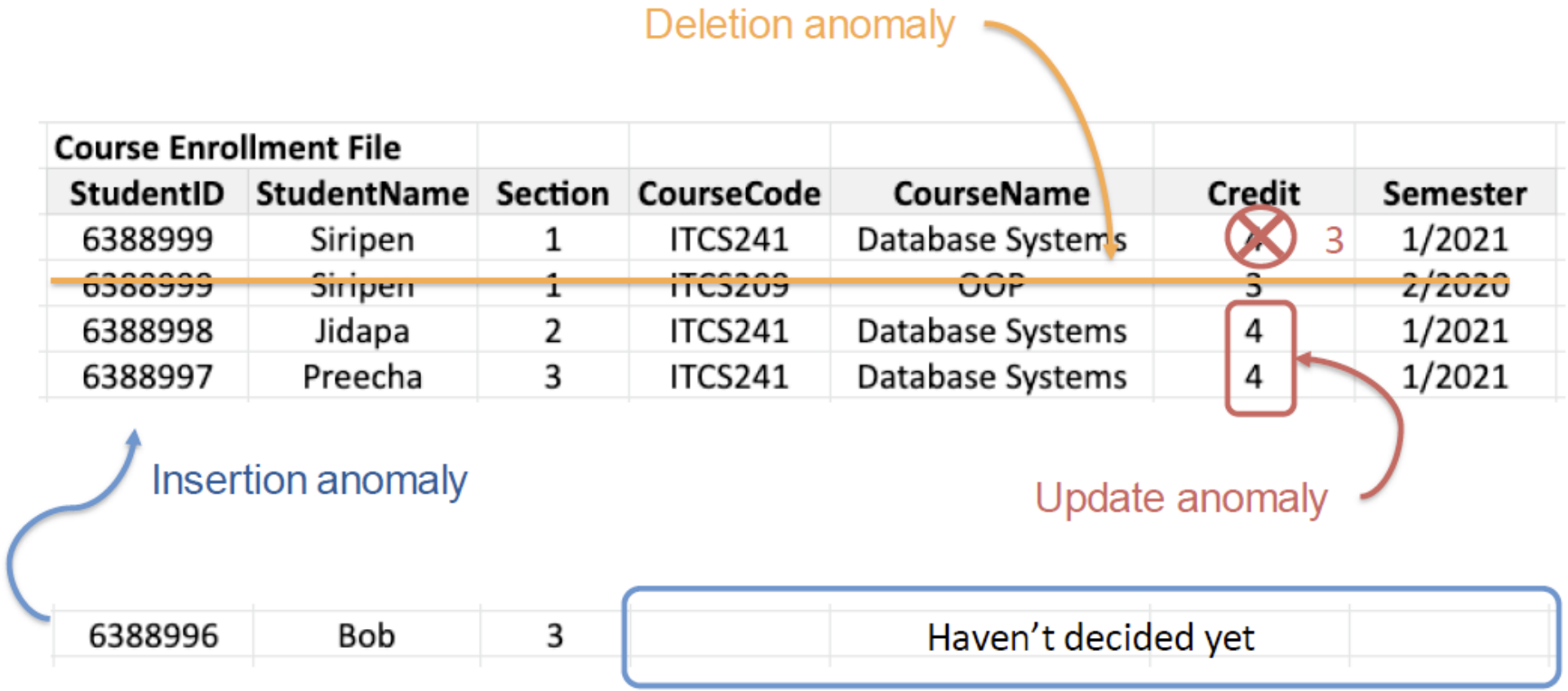
- Data inconsistency: different and conflicting versions of same data occur at different places

Enrollment File

StudentID	StudentName	CourseCode	CourseName	Credit
6388999	Siripen	ITCS241	Database Systems	3
6388998	Jidapa	ITCS241	Database Systems	3
6388997	Preecha	ITCS241	Database Systems	2

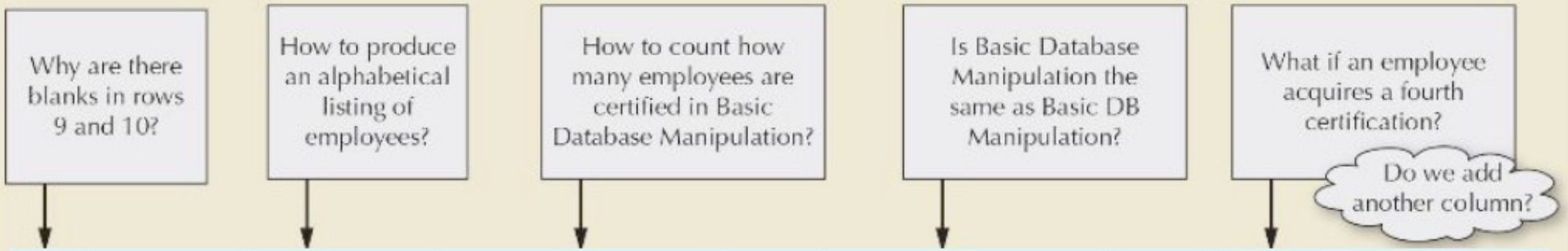
- Data anomalies: abnormalities when all changes in redundant data are not made correctly
  - Update anomalies
  - Insertion anomalies
  - Deletion anomalies

Data Redundancy (cont'd.)



Database Design is Important

FIGURE 1.4 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN



ID	EmployeeNumber	EmployeeName	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
1	02345	Johnny Jones	Basic Database Management	2/14/1993	Advanced Database Management	2/14/1993	Basic Web Design	8/9/1999
2	08273	Marco Bienz	Basic Web Design	3/8/2005	Advanced Process Modeling	8/19/2008		
3	06234	Jasmine Patel	Basic Web Design	8/10/2003	Advanced C# programming	8/10/200	Basic DB manipulation	1/29/2008
4	03373	Franklin Johnson, Jr.	Advanced Spreadsheets	6/20/2007				
5	13567	Almond, Robert	Basic Process Modeling	9/30/2010	Basic Database Design	5/23/2011		
6	10282	Richardson, Amanda						
7	09382	Jessica Johnson	Basic DB Design	8/2/2008	Basic Database Manipulation	8/2/2008	Advanced DB Manipulation	5/1/2009
8	14311	Duong, Lee	Basic Web Design	9/1/2012				
9			Master Database Programming					
10			Basic Spreadsheets					
11	09002	Ben Joiner	Advanced Spreadsheets	5/16/2009	Basic Web Design	5/16/2009		
12	13383	Raymond F. Matthews	Basic C# Programming	3/12/2010				
13	09283	Chavez, Juan						
14	04893	Patricia Richards	Advanced Database Management	6/11/2002	Advanced Database Manipulation	9/20/2008		
15	13932	Lee, Megan						





Database Management System (DBMS)

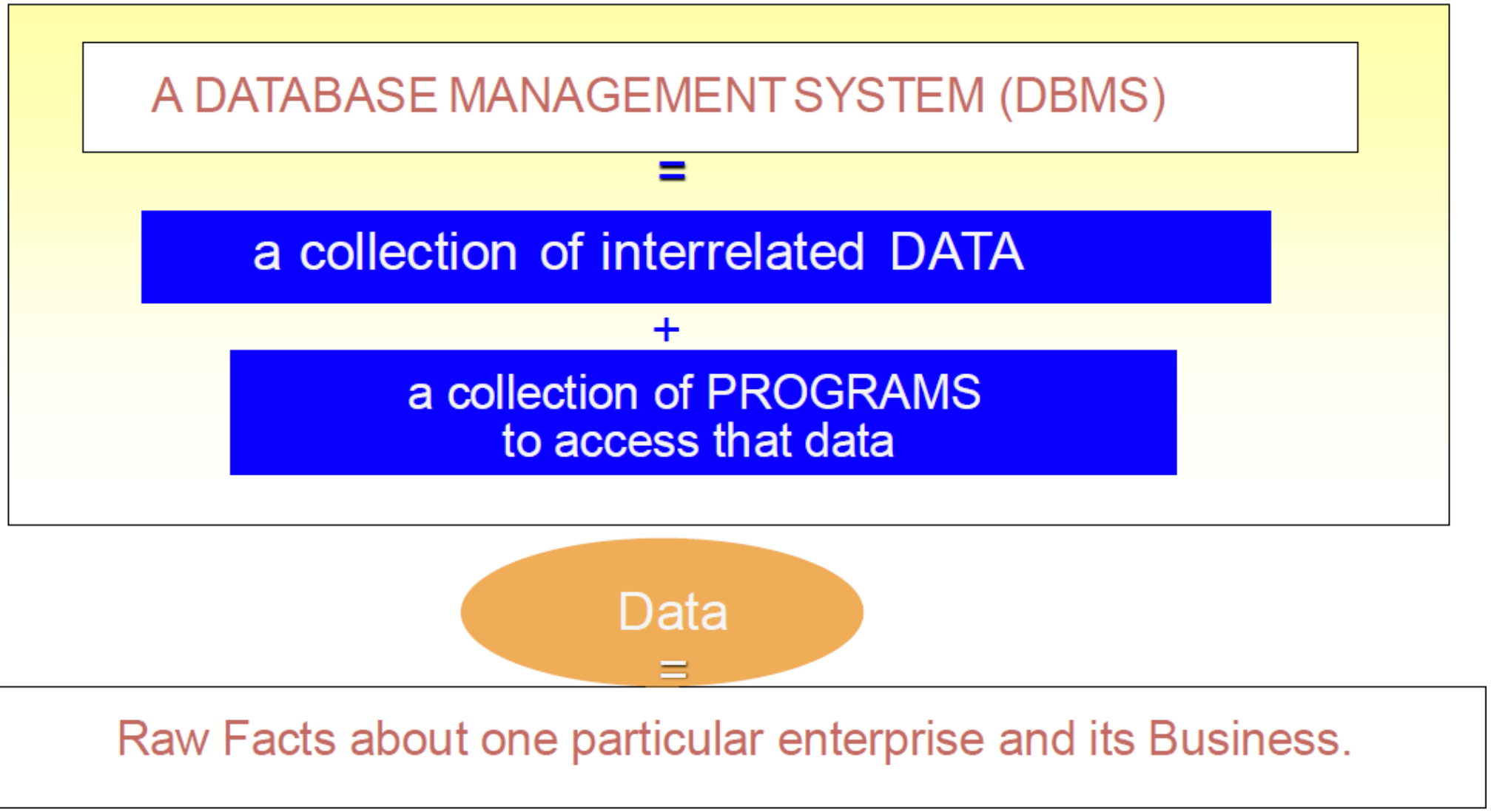
Introducing the Database

- Database: shared, integrated computer structure that stores a collection of:
  - End-user data: raw facts of interest to end user
  - Metadata: data about data
    - Provides description of data characteristics and relationships in data

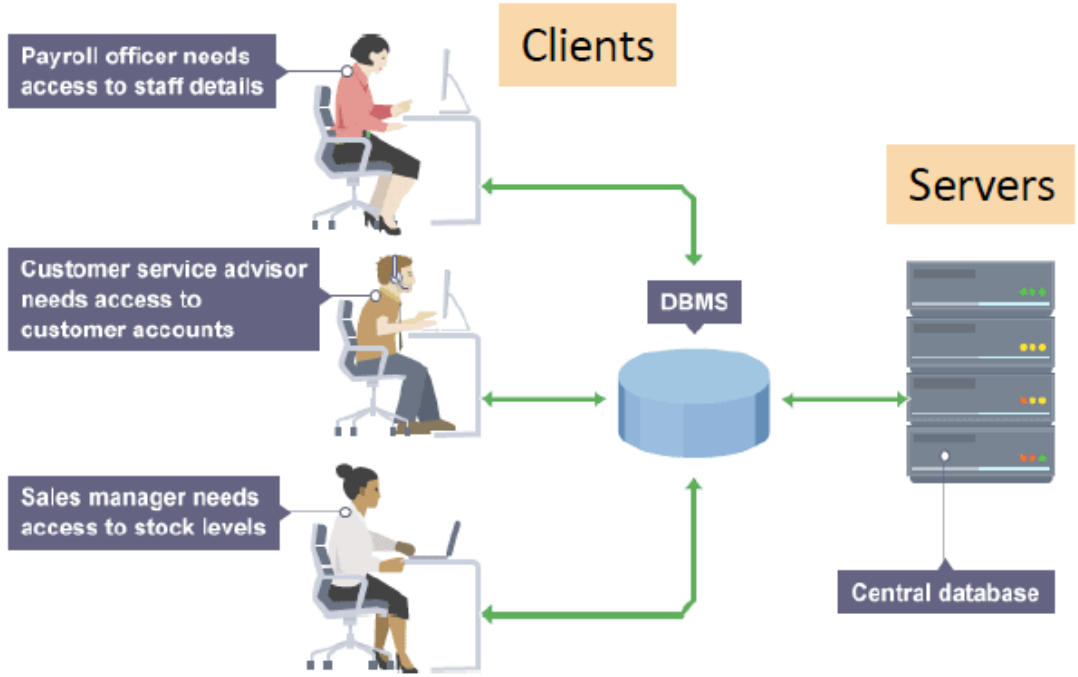


- Database management system (DBMS): collection of programs to
  - Manage the database structure
  - Store the actual data in the database
  - Secure and control access to database



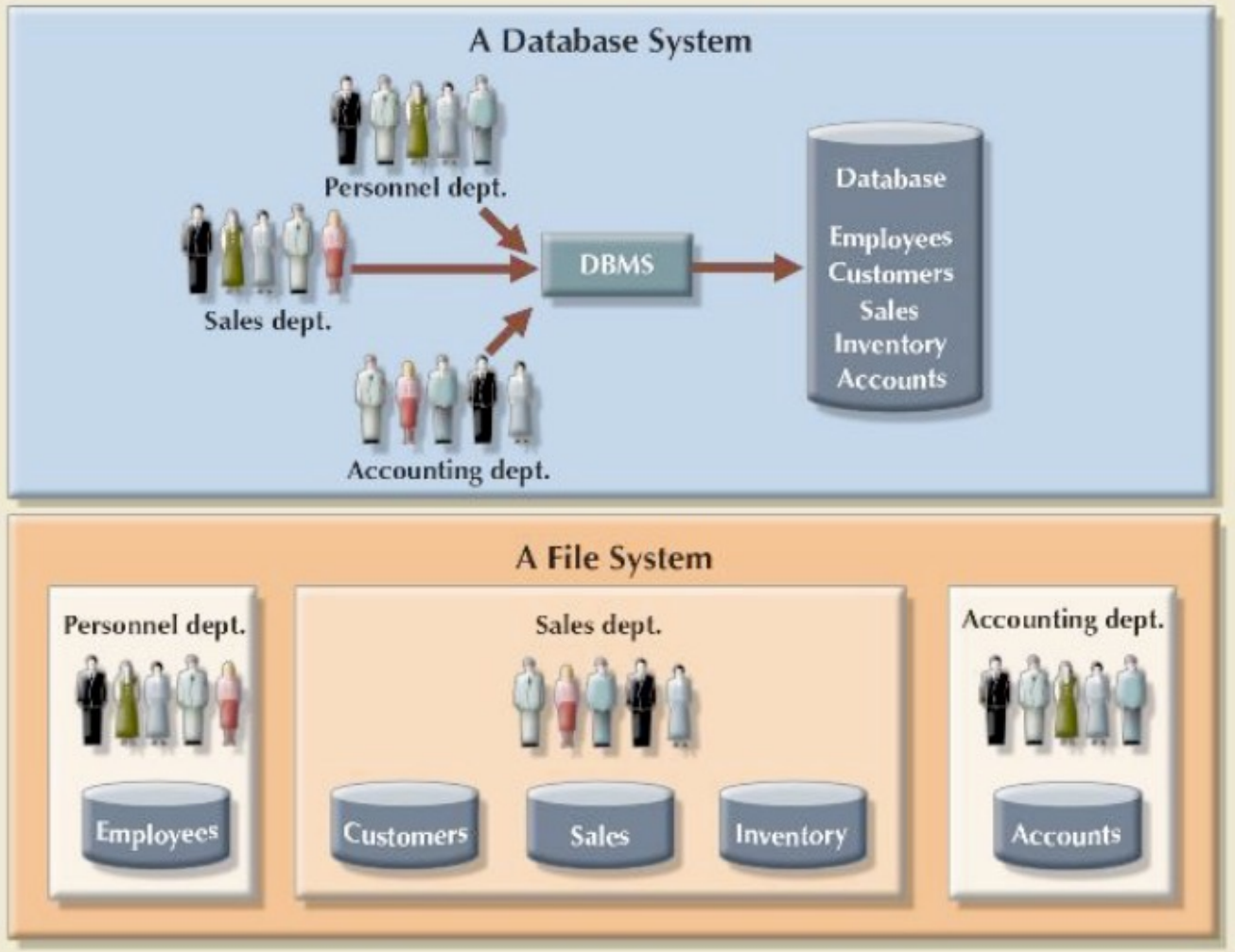

A DATABASE MANAGEMENT SYSTEM (DBMS)
=
a collection of interrelated DATA
+
a collection of PROGRAMS
to access that data
Data
=
Raw Facts about one particular enterprise and its Business.

Role and Advantages of the DBMS



- DBMS is the intermediary between the user and the database
  - Database structure stored as file collection
  - Can only access files through the DBMS
- DBMS enables data to be shared
- DBMS integrates many users' views of the data

FIGURE 1.9 CONTRASTING DATABASE AND FILE SYSTEMS

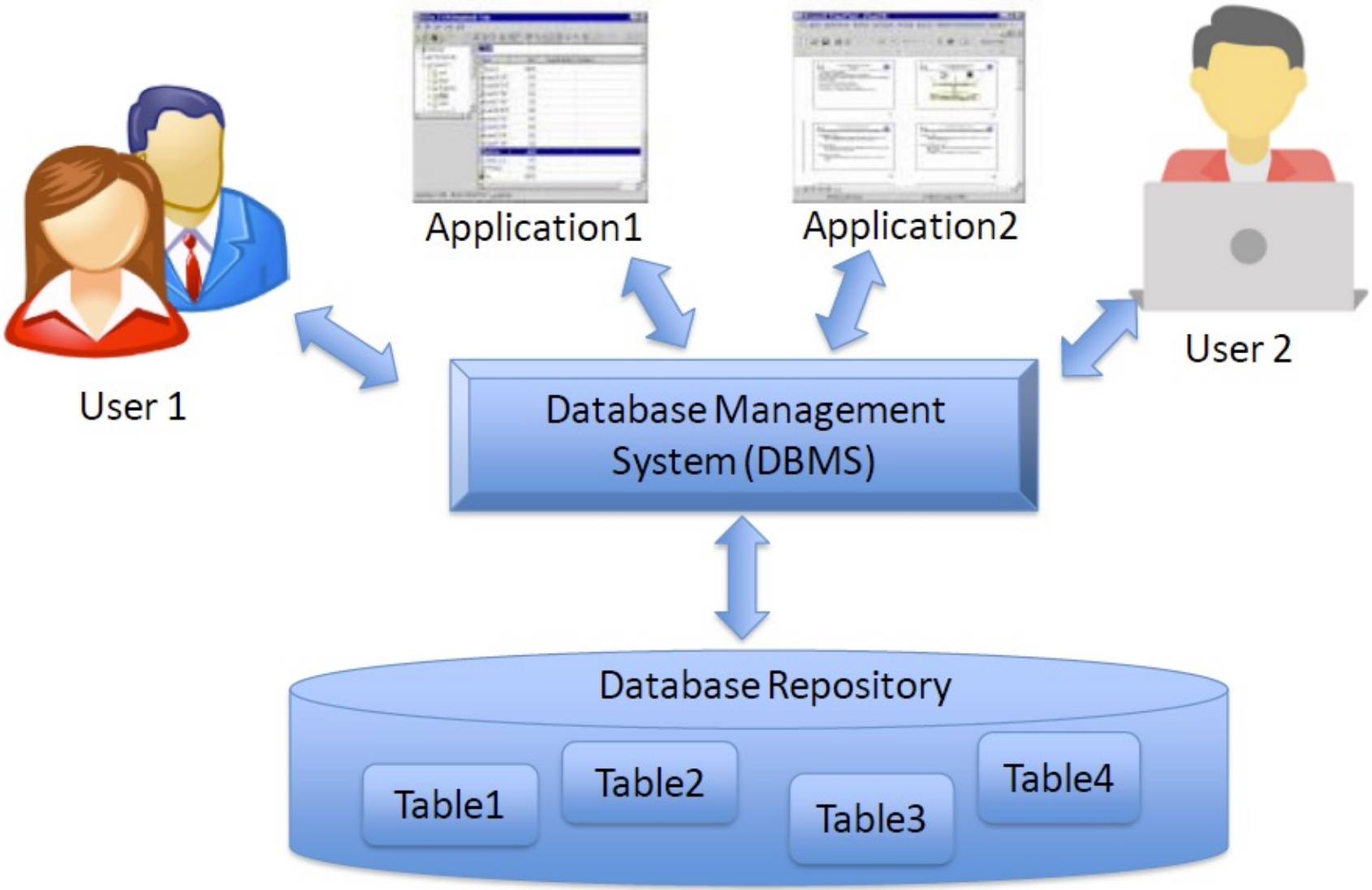




Database Concept, Architecture and Components

Database Concept

From end-user to database implementation



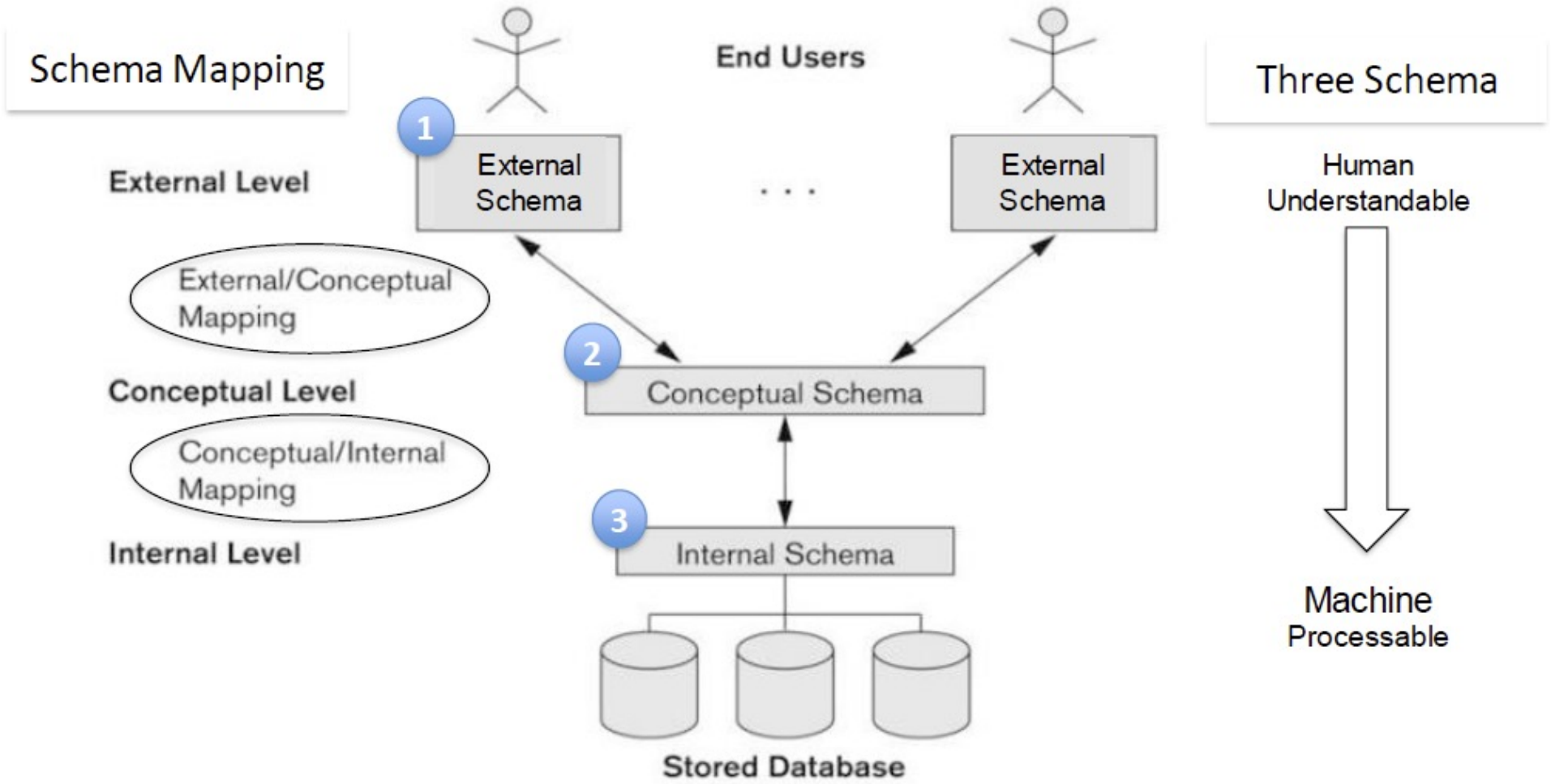
DBMS Functions

- Data storage management
  - DBMS creates and manages complex structures required for data storage
  - Also stores related data entry forms, screen definitions, report definitions, etc.
  - Performance tuning: activities that make the database perform more efficiently
  - DBMS stores the database in multiple physical data files
- Database access languages and application programming interfaces
  - DBMS provides access through a query language
  - Query language is a nonprocedural language
  - Structured Query Language (SQL) is the de facto query language
    - Standard supported by majority of DBMS vendors
- Data integrity management
  - DBMS promotes and enforces integrity rules
    - Minimizes redundancy
    - Maximizes consistency
  - Data relationships stored in data dictionary used to enforce data integrity
  - Integrity is especially important in transaction-oriented database systems

Database Concept

Recommend

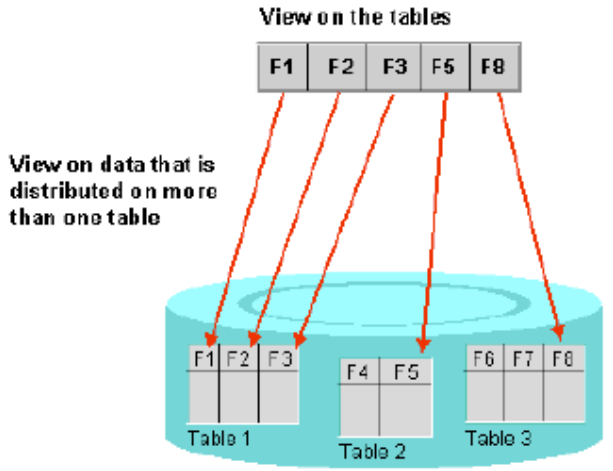
Three Schema Architecture

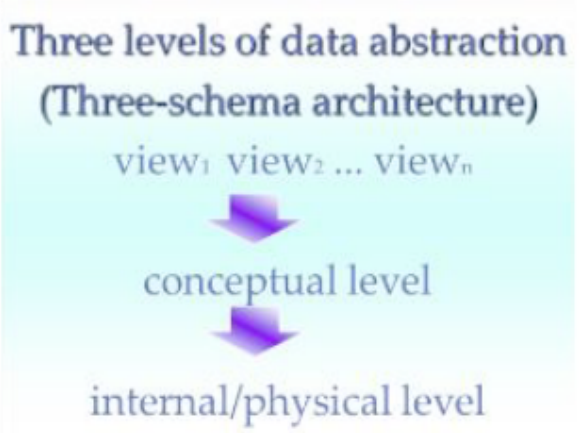


1. EXTERNAL VIEW OR EXTERNAL SCHEMA

(what are actually seen by the users)

- A highest level of abstraction describes a part of the entire database of the specific user's interest.
- VDL (view definition language) is a facility for declaring views for the users.
- VML (view manipulation language) is a facility for expressing queries and operations on the views.
- Most DBMS has VDL & VML as a part of its SQL.



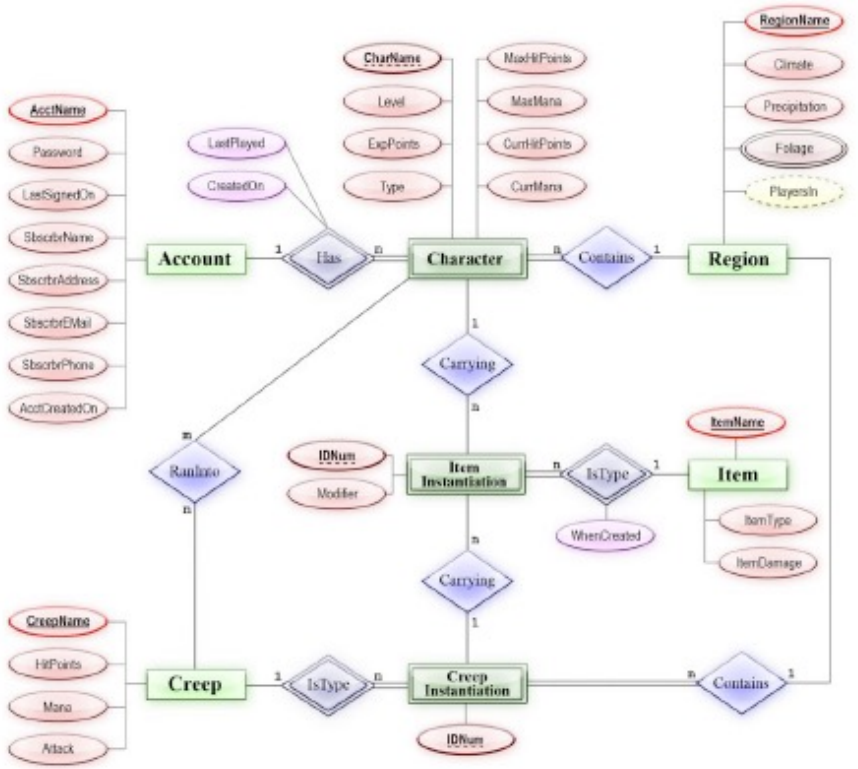


2. Conceptual Schema

(what data are actually conceptualized)

A **conceptual schema** is a representation of informational needs underlying the design of a database.

- It describes the structure of the whole database.
- Conceptual schema hides the details of physical storage structures and emphasizes on describing entities, data type, relationship & operations.

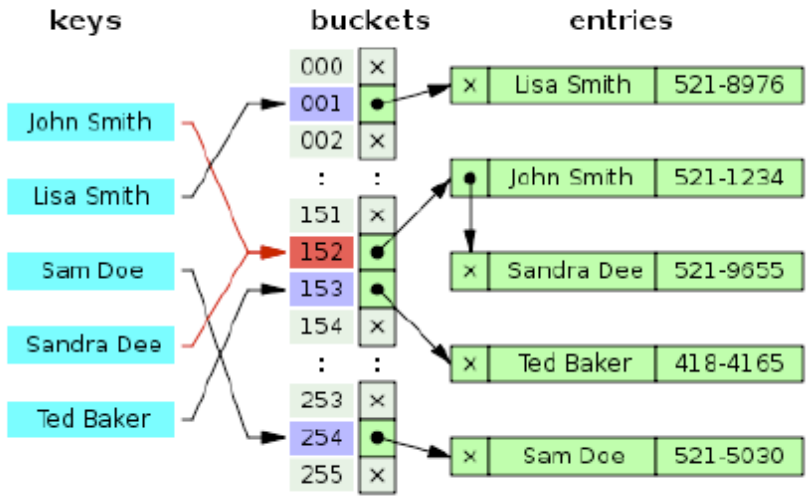


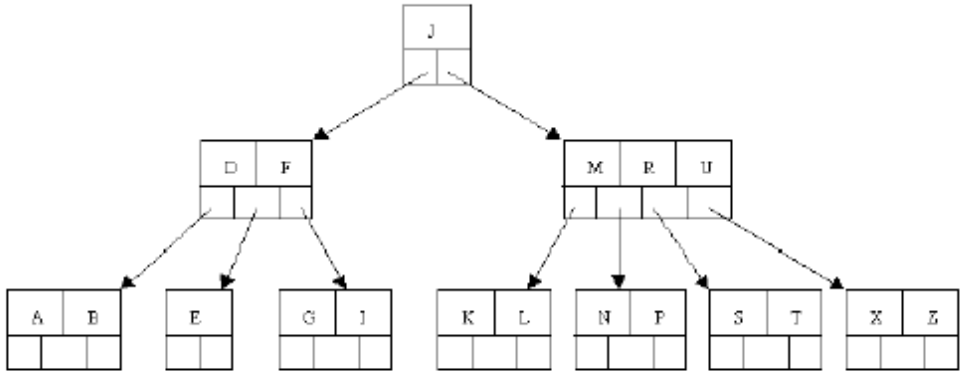
3. Internal/Physical Schema

(how the data are stored)

Physical schema is a term used in data management to describe how data is to be represented and stored (files, indices, etc.) in secondary storage using a particular database management system (DBMS).

Storage media such as Cache, Main Memory, Disk, Tape
Access methods : Direct access & Sequential access





1

External Schema or External View for each user

Schema Mapping

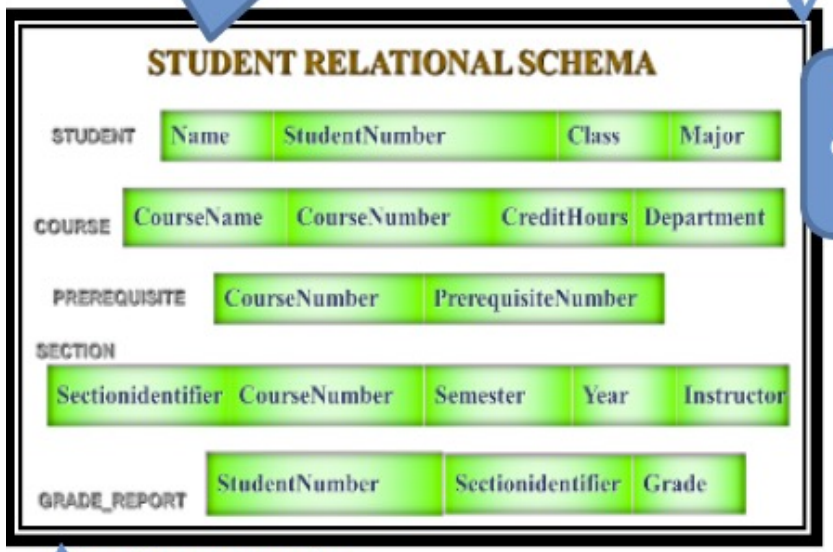
2

Conceptual Schema or Logical Schema for each Database Application

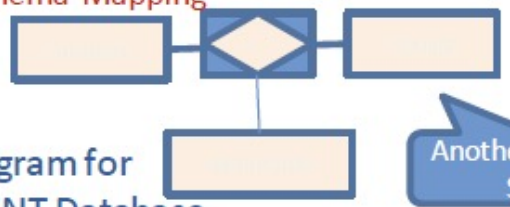
Schema Mapping

Physical Schema for each Database stored in the disk storage

3



Schema Mapping



ER Diagram for STUDENT Database

Another Conceptual Schema

a) The student transcript view

VIEWS

TRANSCRIPT

StudentName	CourseNumber	Grade	Semester	Year	SectionId
Smith	COSC1310	C	Fall	92	119
	MATH2410	B	Fall	92	112
	MATH 2410	A	Spring	91	85
Brown	COSC1310	A	Fall	91	92
	COSC3320	B	Fall	92	102
	COSC3380	A	Fall	92	135

StudentTranscript

Example of a database: Views

b)The course prerequisite view

PREREQUISITES

CourseName	CourseNumber	Prerequisites
Database	3380	COSC3320
		MATH2410
Data Structure	3320	COSC1310

Three Levels of Data Abstraction - - a characteristics of Data and Program independency

Storage format for a STUDENT record.

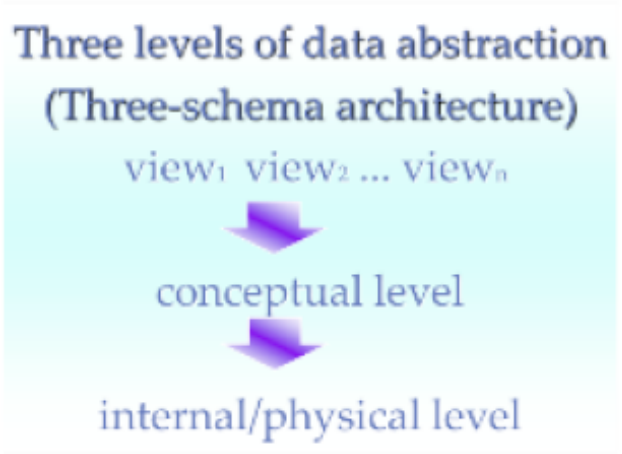
STUDENT	Name	StudentNumber	Class	Major
	Smith	17	1	COSC
	Brown	8	2	COSC

Data Item Name	Starting Position in Record	Length in Characters (bytes)
Name	1	30
StudentNumber	31	4
Class	35	4
Major	39	4

Benefit of Three Shema Architecture

Three SCHEMA Levels make DATA and Structural INDEPENDENCE happened

- ✓ An ability to modify a scheme in one *lower level* without affecting a scheme in the next *higher level*.



- ✓ There are two levels of independence:
  - **Physical independence** (modify physical schema NOT affect Conceptual Schema)
  - **Logical independence** (modify Conceptual schema NOT affect External Schema or Application Program)

More to Read

- Chapter 1: Database Systems from "***Database Systems: Design, Implementation, & Management***" **13th Edition** by Carlos Coronel and Steven Morris